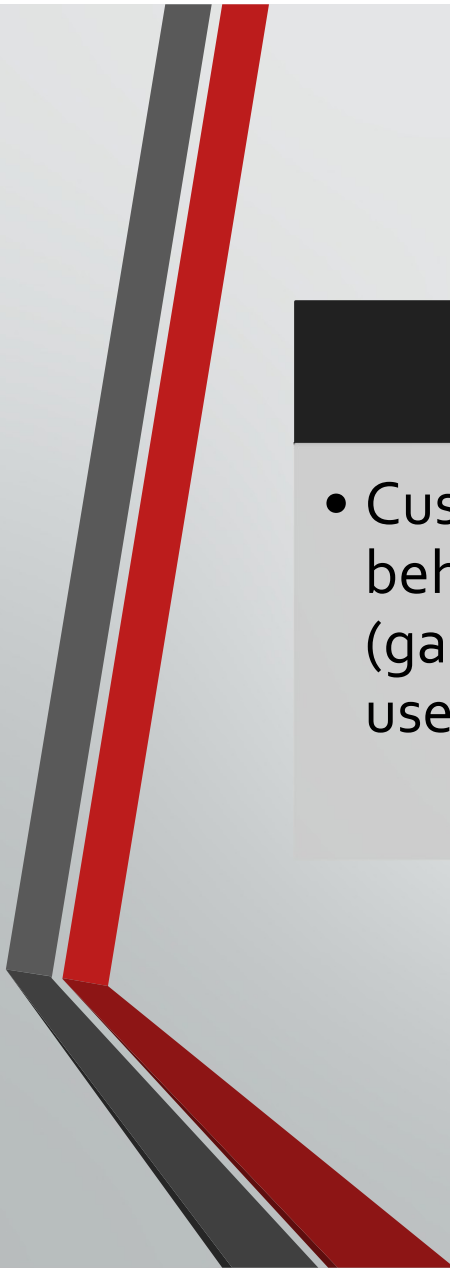




Customer Loyalty Analysis: Drivers and Customer Segmentation

By
Lakmali Amarawardana



Dataset

- Customer behaviour (gaming/platform users)

Objective

- Understand drivers of loyalty points and segment customers

Methods

- Correlation Analysis, Regression, Decision Tree, K-Means clustering



Objectives

- Identify key drivers of **loyalty points**
- Understand relationship between:
 - Income
 - Spending behaviour
 - Age, education, product
- Build **predictive models**
- Create **customer segments for marketing**

Correlations

Strong relationships

- Spending score - Loyalty points (**0.67**) 🔥
- Income - Loyalty points (**0.62**)

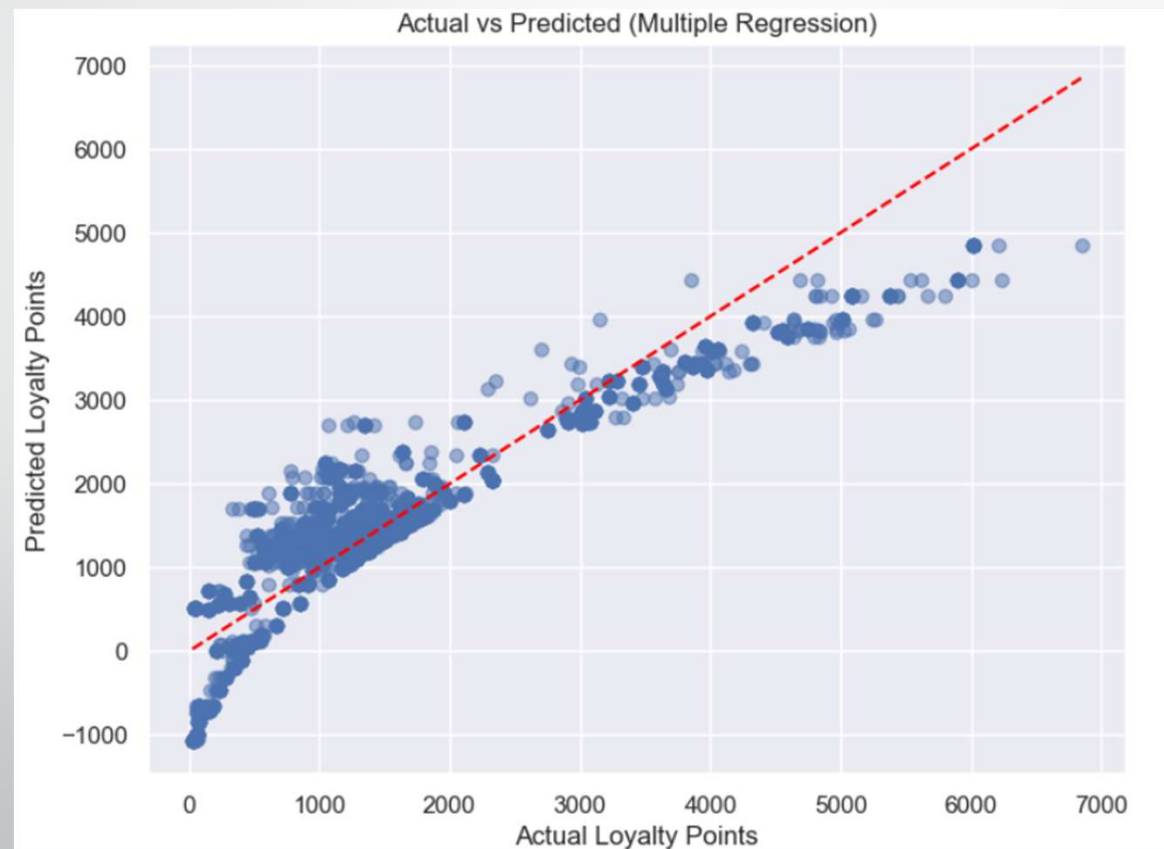
Weak relationships

- Product - Loyalty points (**0.18**)
- Age - Loyalty points (**-0.04**)

Insight

Loyalty drivers **behaviour + income**, not demographics.

Multiple Linear Regression



Multiple Linear Regression

Model	Results	Coefficients	Insight
• Target: Loyalty_points • Predictors: income + spending_score	• $R^2 = 0.827$ • F-statistic • p-value = 0.00 (highly significant)	• Income = 33.98 • Spending_score = 32.89	• Both income and spending significantly contribute to loyalty prediction.

Regression Diagnostics



Residuals

Non-linearity
present

Heteroscedasticity
present



Normality

Slight skew,
acceptable
range



Insight

Model
assumptions are
partially satisfied



Decision Tree Model

Stats

$R^2 = 0.882$

RMSE = 440.45

Feature importance

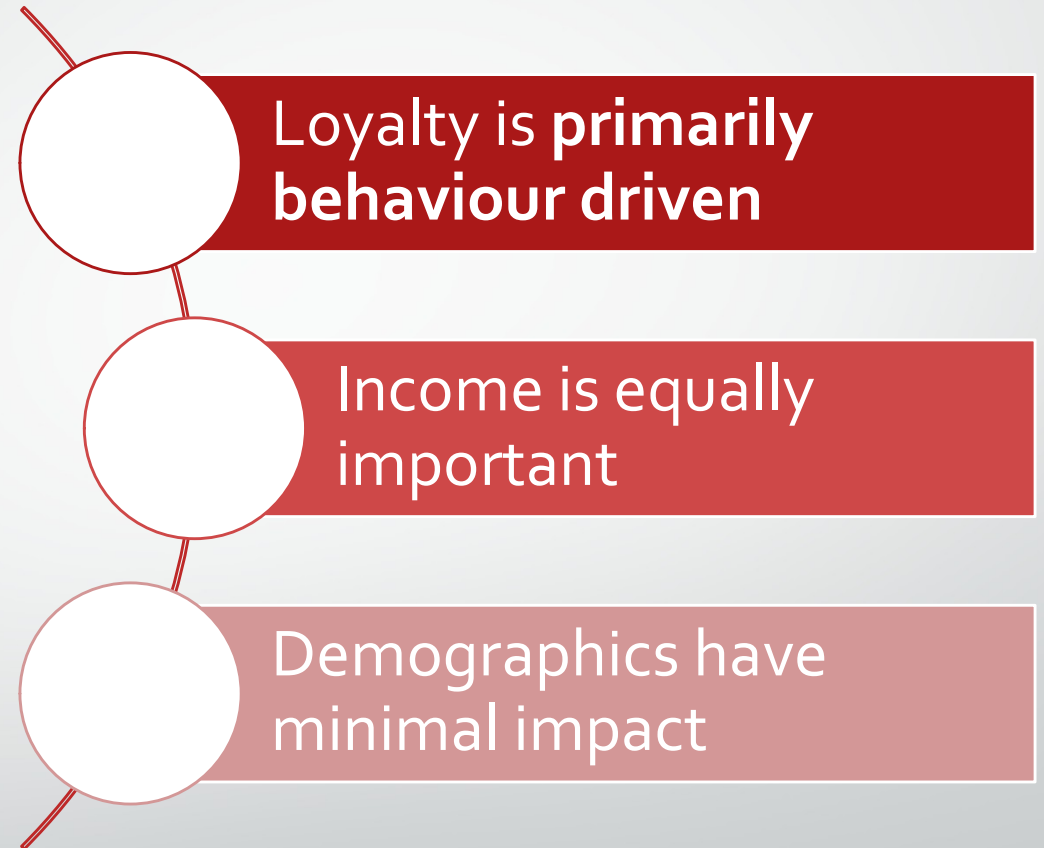
Spending → 0.515

Income → 0.483

Insight

Behavioural features
dominate prediction
(~99%)

Key Business Insight



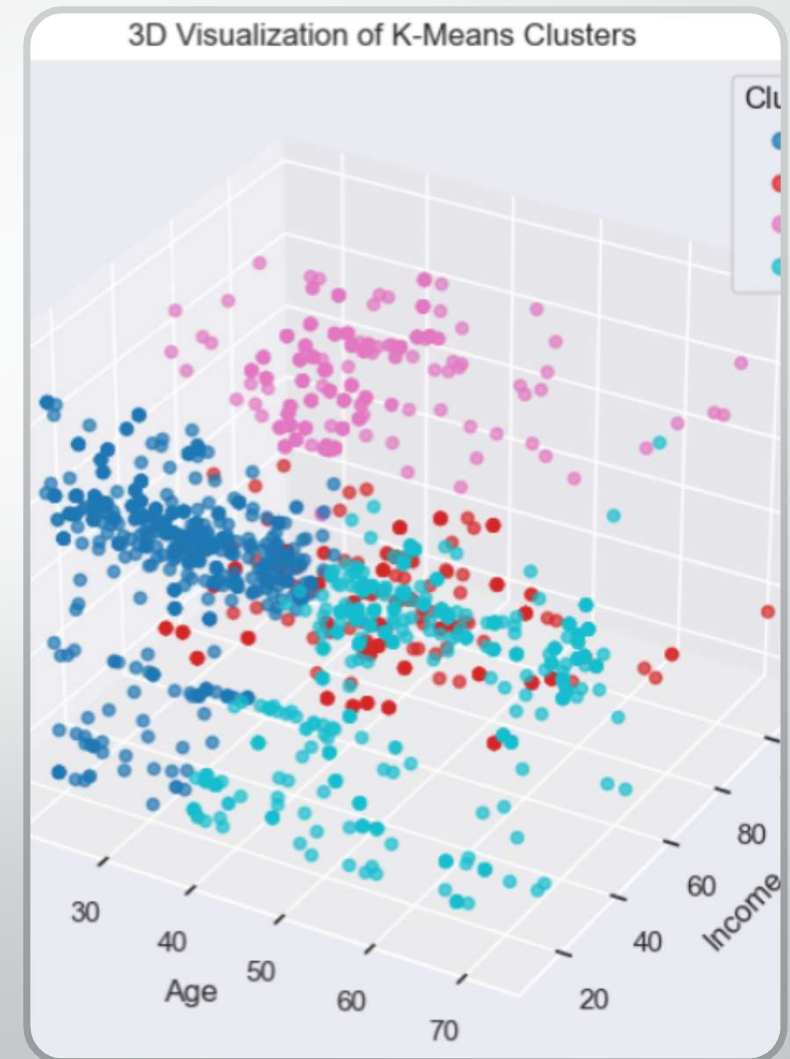
Clustering (K-Means)

**k=7 provides better granularity
than k=5**

Identifies niche behavioural groups

Example clusters:

- **VIP customers**
- **Low value users**
- **Impulsive low - income
Spenders**
- **Wealthy but low engagement**

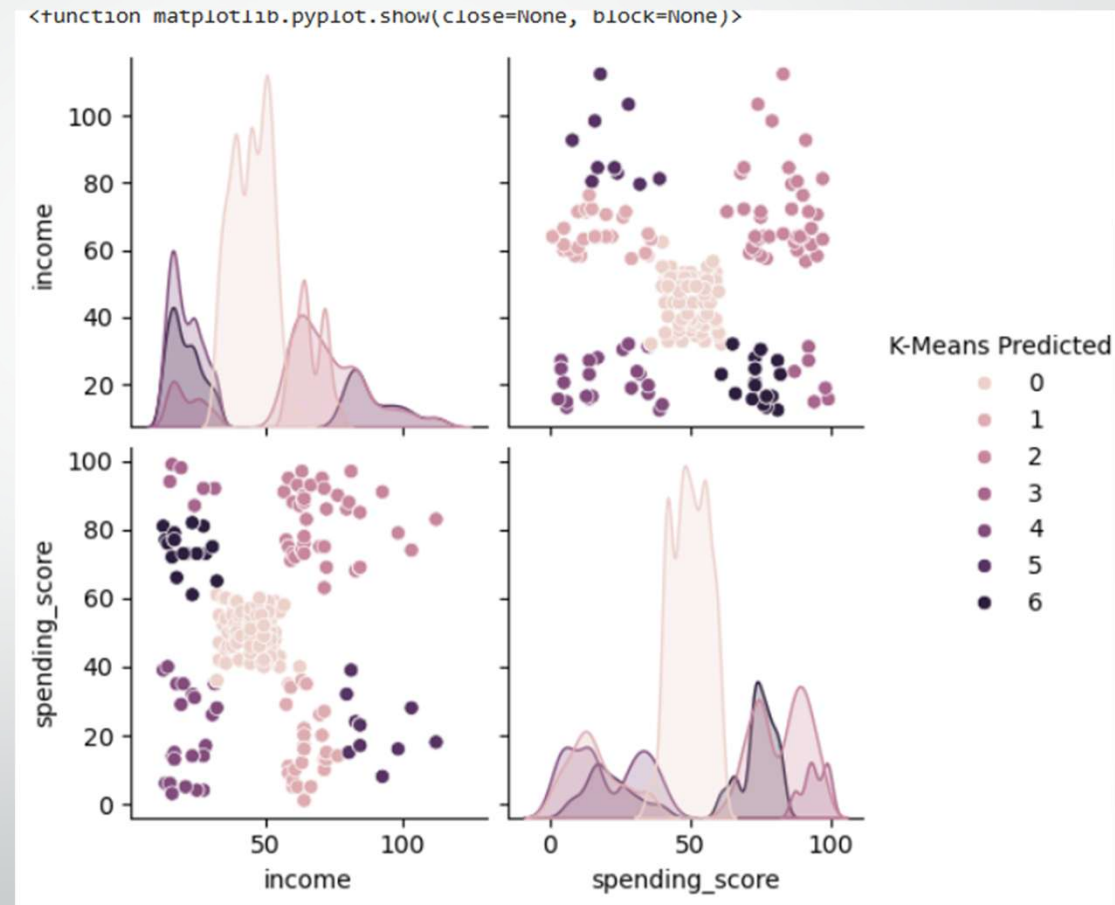


Cluster Behaviour Insight

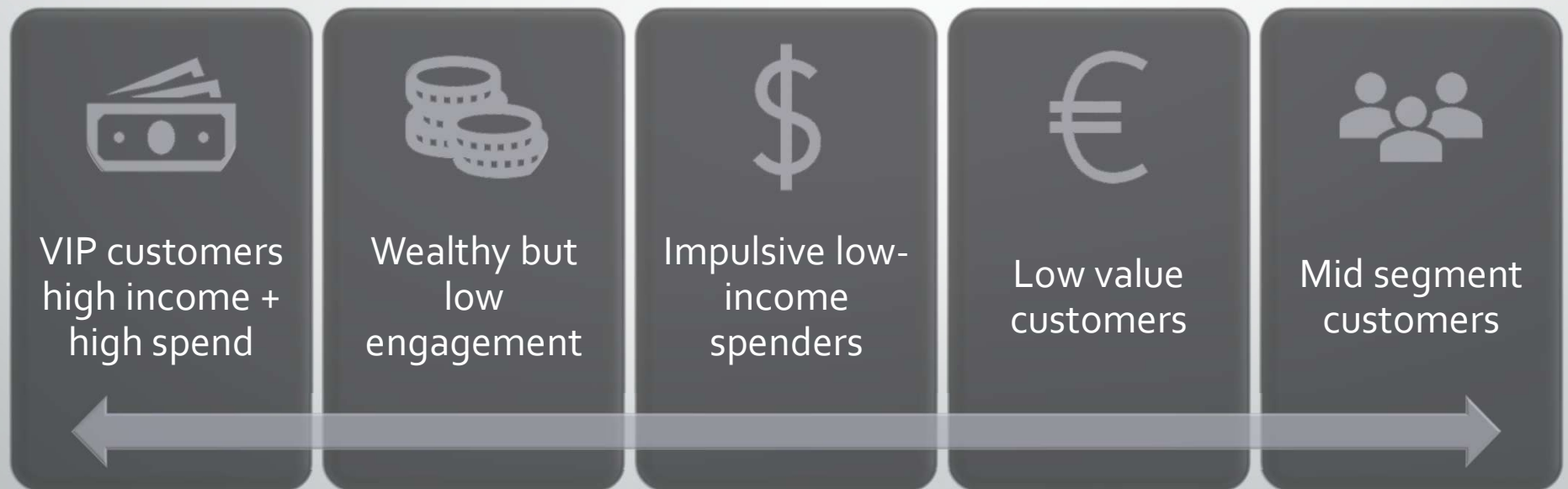
Even with similar income (12–13 range):

Cluster 4 - Low engagement users

Cluster 6 - High engagement users



Final customer segments



Conclusion



Loyalty is driven by:

- Spending behaviour (strongest)
- Income (secondary)



Weak influence:

- Age
- Education
- Product

Message



Customer loyalty
is primarily
**behavioural, not
demographic**

Recommendations

Target VIP
customers with
rewards

Re-engage
wealthy low-
spenders

Design offers for
impulsive low-
income users

Reduce marketing
spend on low-
value segment



A mini story!

Hard Cluster Boundaries

Customer Segments are Not Always Clear-Cut

Cluster assignment
of customers

**Hard (not
probabilistic)**

Located close to
boundary

Some customers sit
**very close to
multiple
segments**

Problem

Creates uncertainty
at the segment
boundaries

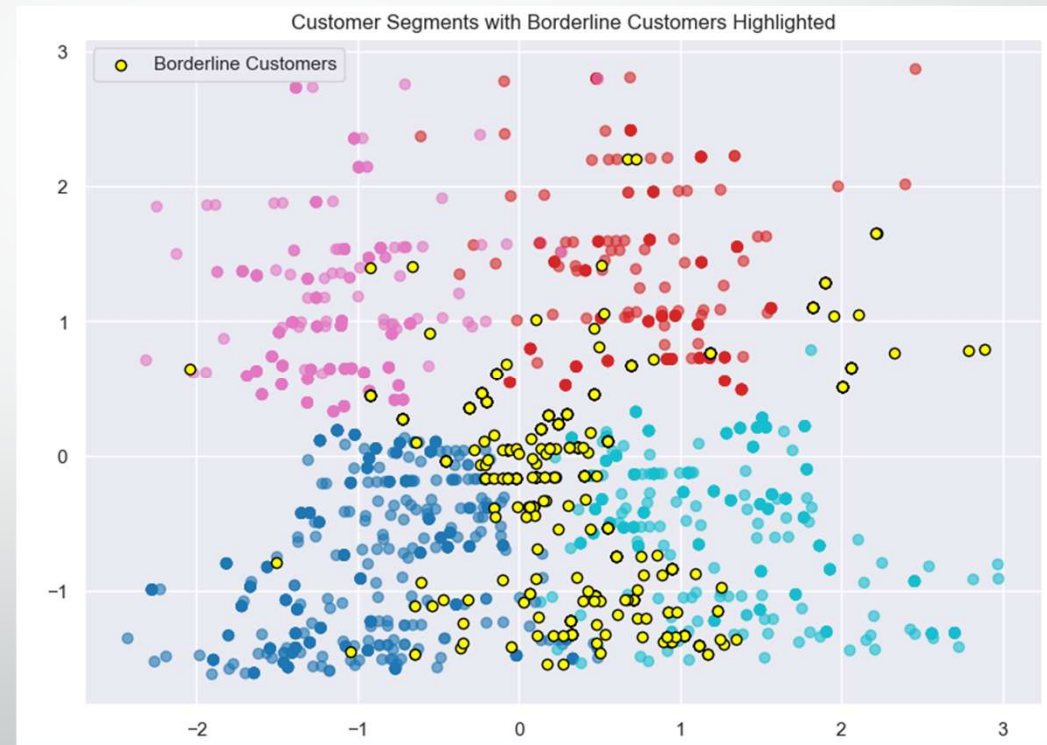
 Key issue

- Not all customers clearly belong to one segment

Borderline Customers: Segment Uncertainty

~400 customers are
borderline cases

Their assignment is
not strongly distinct
to one segment



Treat as a monitoring group

